iMouse 2.0

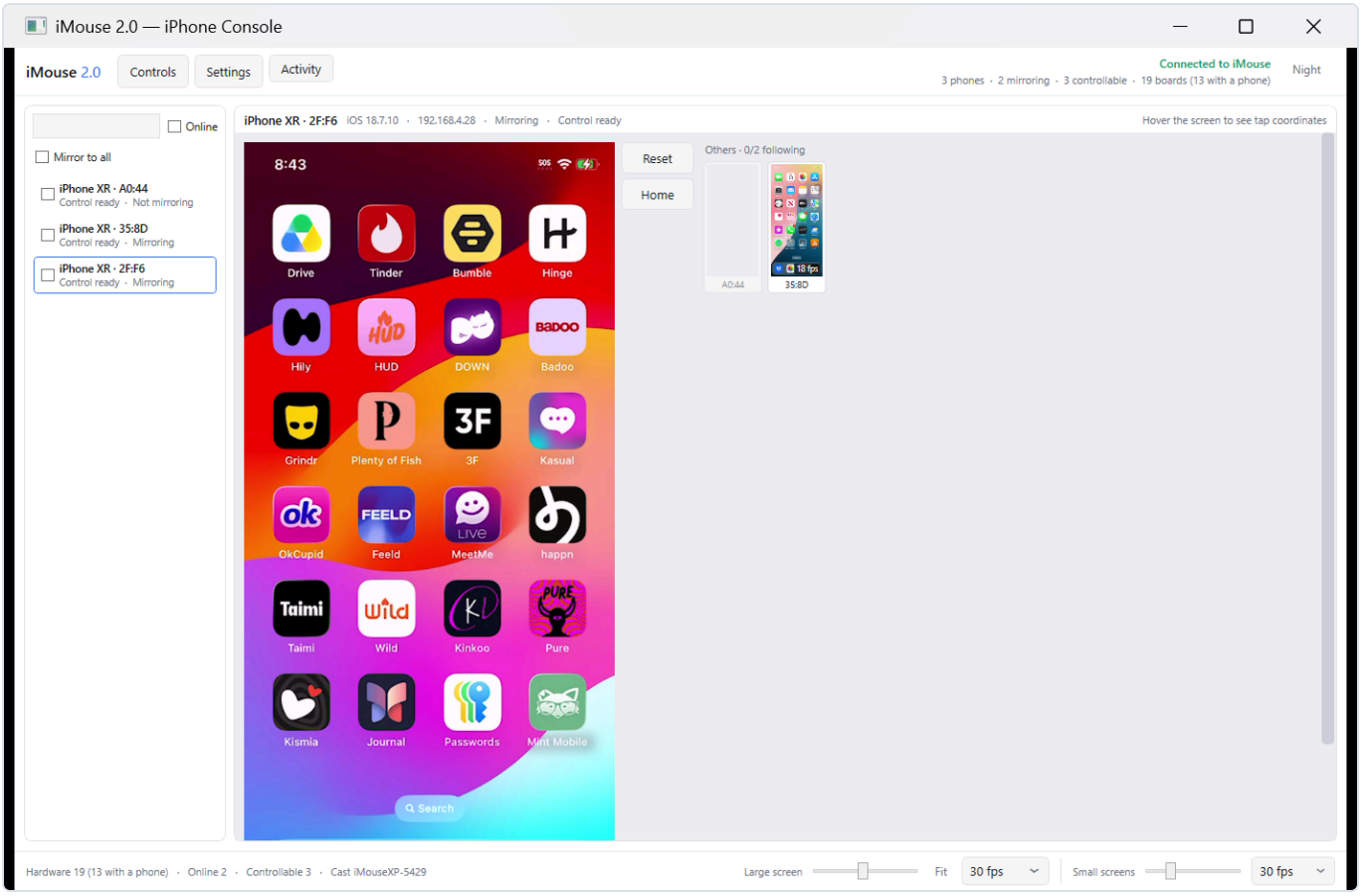
This window shows a real iPhone's screen and turns your mouse clicks into real taps on that phone. This manual tells a screenshot-driven agent exactly what is on the screen, what every control does, how to verify each action landed, and which clicks are dangerous.

1 click on the phone image = 1 real tap

Mirror to all = that tap on every phone

ASCII only, via Send text

~1 s then screenshot to verify



Console build 2.0.0, XP kernel on 127.0.0.1:9911, display 2560×1440 at 150% scaling

Read section 1 before acting

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SECTION 1 · ORIENTATION

Read this first

You are an agent that sees screenshots of a Windows desktop and acts with mouse and keyboard. The window you are operating is not an ordinary app. The large picture inside it is the live screen of a real iPhone, and clicking on that picture sends a real tap to that phone. Everything in this manual follows from that.

What iMouse 2.0 is

iMouse 2.0 is a Windows program that controls a rack of physical iPhones. Each phone is wired to a small control board that the phone believes is a keyboard and mouse. Each phone also streams its screen to the PC over Wi-Fi. The console puts one phone's live screen in the middle of the window, the other phones as small tiles beside it, and a list of all phones on the left. A click on the large screen becomes a tap at that spot on the phone. A drag becomes a swipe. Buttons around the picture send Home, Back, swipes and typed text.

There are two separate paths for each phone, and they fail separately. **Control** is the wired path: PC to board to phone. **Video** is the wireless path: phone to PC. A phone can show a perfect picture and ignore every tap, or accept every tap while showing nothing. The console always reports both, and so must you.

The five rules

1 **A click on the phone picture is a real tap on a real phone.** Never click inside the large phone image unless a tap at that spot is the action you intend. Never click there to "focus" the window, to dismiss something, or by accident while moving toward a button.

2 **Check how many phones you are about to hit before every phone action.** With mirroring on, one click or button press goes to every checked phone at once. The count is printed on screen: a green `→ N phones` under the Reset and Home buttons, and `Master: … → mirroring to N phone(s)` in the Controls drawer. If you did not intend N phones, untick **Mirror to all** first.

3 **Your keyboard types into the console, not into the phone.** A `type` action goes to whatever Windows control has focus. The only way to type on the phone is the **Send text** box in the Controls drawer followed by the **Send text** button, and it accepts plain English letters, digits and punctuation only.

4 **Act, wait about one second, screenshot, verify.** The phone picture is live, so the result of a tap appears in your next screenshot. Never chain phone actions blind. If the picture did not change, follow the recovery steps in Section 7 instead of repeating the action.

5 **Moving the mouse over the phone picture moves the phone's own pointer.** Hovering is harmless but visible: a small grey dot follows your cursor on the phone. Park the cursor on a neutral area (the dark stage around the picture, or the status bar) when you are not acting.

How to use this manual

- Section 2 is for whoever wires you up: screen size, scaling, window placement, and how your action vocabulary maps onto this window.
- Sections 3 and 4 describe every control, where it is, and what changes on screen when it is used. Exact on-screen strings are shown like `this`. Control labels are shown like **this**.
- Section 5 lists the strings that tell you the state of the connection, the phones and the mirror set. Read them before acting, not after.
- Section 6 gives step-by-step procedures. Each step says what you should expect to see next. If you do not see it, stop and go to Section 7.
- Section 8 is the list of clicks that cause harm. Section 9 is a one-page reference.

The rig behind the window

(the vendor kernel, IMService.exe, which must already be running)

Fact	Value on this PC
Program	`C:\Users\joey_\Desktop\Claude Code\iMouseXP-2.0\release\iMouse2\iMouse2.exe`
Window title	`iMouse 2.0 — iPhone Console`
Default window size	1420 × 940 logical pixels; minimum 980 × 640; opens centred
Display	2560 × 1440 physical pixels at 150% scaling, so the window is about 2130 × 1410 physical pixels
Service it talks to	`http://127.0.0.1:9911`
Phones on the rack	iPhone XR handsets; their mirror image is 500 × 1080 pixels, which is the coordinate space taps use
Theme	Light or night, remembered between runs. The top-right button shows the theme you would switch *to*: `Night` means light is active, `Light` means night is active

SECTION 2 · FOR WHOEVER WIRES UP THE AGENT

Setting up the agent and its harness

This section is for the person configuring the computer-use agent, not for the agent mid-task. It covers grants, screen geometry, and the one rule that keeps a screenshot agent safe on this window: read your own screen, never trust coordinates copied from anywhere.

What the agent needs access to

- **One application only:** the window titled `iMouse 2.0 — iPhone Console`. Grant that and nothing else. In particular do not grant the vendor windows (IMConsole, IMService) or let the agent roam the desktop.
- **No system key combos** are needed. The agent never needs to quit or switch apps. Withhold that grant.
- **Clipboard** is not needed for normal operation. The console's **Paste** button pastes the phone's own clipboard, not the PC's.

Screen geometry on this PC

Property	Value	Why the agent cares
Display	2560 × 1440, 150% scaling	The agent's screenshots arrive at the scaled size. It must click by what it sees, not by fixed numbers.
Console window	1420 × 940 logical, opens centred, resizable	The window can be moved or resized between runs, so absolute positions are never safe to reuse.
Phone image inside it	Sized by the **Large screen** slider, default about 620 px tall	The mapping from a screen pixel to a phone tap is done by the console. The agent just clicks the spot it wants.

! **The figures in this manual are illustrations, not a coordinate source.** They were captured at 2130 × 1410 in the night theme. Your screenshots will differ in size, theme and phone content. Every number in this manual (a percentage, a pixel) describes where a control sits *relative to the window* so you can find it in your own screenshot. Never click a raw coordinate from this document. Locate the control by its label and its region, then click what you see.

The loop the agent should run

1. **Screenshot.** Look at the whole window first.
2. **Read state** (Section 5). Is it connected? Which phone is selected? How many phones will an action hit? Is the phone controllable, or only visible?
3. **Decide the single next action.** One tap, one button, one typed string.
4. **Confirm the blast radius.** If the action touches the phone and the mirror count is not what you intend, fix the mirror set first (Section 6).
5. **Act.**

6. **Wait about one second** (two on a slow phone), then screenshot and verify the expected change (Section 7). Do not issue the next phone action until you have verified this one.

Preconditions before the agent starts

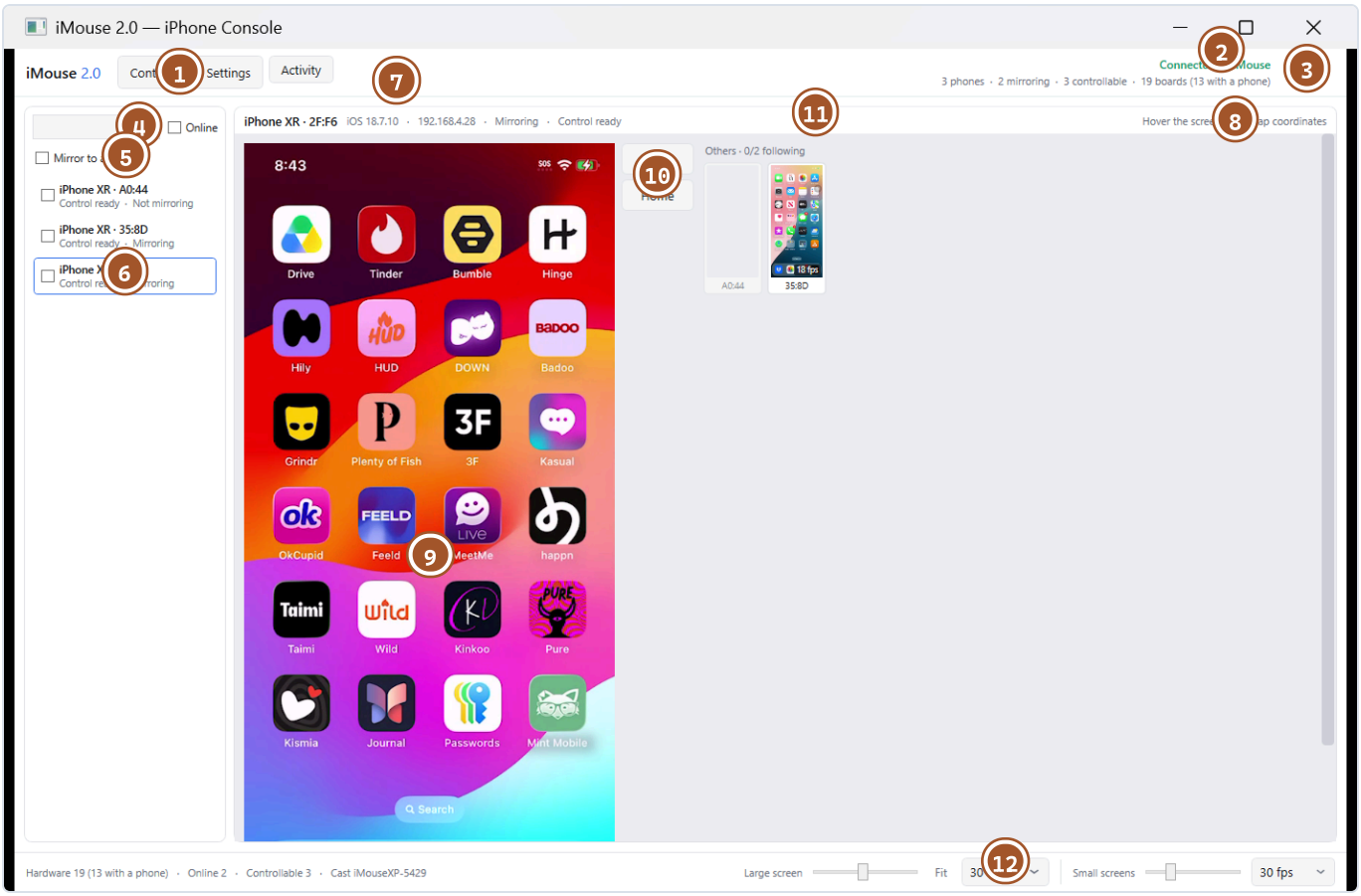
- The vendor kernel (`IMService.exe`) is running, and ideally the vendor console (`IMConsole.exe`) too, because the phones cast to it. If phones show `Not mirroring` , a person may need to start the vendor console first.
- The console shows `Connected to iMouse` at the top right. If it shows `Service unavailable · retrying` , the agent cannot do anything useful; stop and raise it.
- At least one phone in the left list shows `Control ready` in green. A phone that is only `Mirroring` can be seen but not driven.

A safe first action to prove the loop

Select a phone, then press **Home** in the strip beside the phone image, with the mirror set empty (no green `→ N phones` line). The phone should return to its home screen within a second, and the activity log should show `Home` . If that works, the click-wait-verify loop is functioning. If the picture does not change, go straight to Section 7 rather than repeating.

The screen, region by region

The window has four bands: a top bar, a left device list, a large phone stage in the middle with a wall of small phones beside it, and a bottom status bar. This is the whole surface while driving. The command buttons live in a drawer that slides in from the right, covered in Section 4.



The main window, night theme. The large screen is a real iPhone. The numbered regions match the table below. Your window may be lighter, a different size, and show different phone content.

#	Region	What it is and what it does
1	**Controls / Settings / Activity**	Three buttons that open the right-hand drawer to that page. Section 4. Opening a drawer changes nothing on any phone.
2	Connection status	Text only. `Connected to iMouse` when healthy, with a summary line under it: phones, mirroring, controllable, boards. Section 5.
3	**Night** / **Light**	Repaints the console. The label is the theme it will switch *to*. Harmless, but it changes every colour you are matching against.
4	Search box and **Online** filter	Typing here filters the device list by name. It types into the console, not the phone. **Online** hides phones that are not mirroring.

#	Region	What it is and what it does
5	**Mirror to all**	A master switch. Ticking it checks every phone and turns mirroring on, so one action hits the whole rack. Unticking releases them. Its label shows the count: `Mirror to all  (3 of 3)`.
6	Device list	One row per phone. Click a row to make that phone the large one. The checkbox on the left adds it to the mirror set. Each row shows two states in words: control, then video. Section 5.
7	Selected phone header	The name and one detail line for the phone on the stage: `iOS 18.7.10 · 192.168.4.28 · Mirroring · Control ready`.
8	Pointer readout	Tells you where a tap would land as you hover: `x 208, y 640 — click to tap here`, or `Outside the phone screen`. Read this before clicking the picture.
9	The phone stage	**The live phone screen. A click here is a real tap. A drag here is a real swipe.** This is the one dangerous region on the window.
10	Action strip	Two buttons against the phone: **Reset** (re-centre the phone's pointer) and **Home**. If mirroring is on, a green `→ N phones` line appears here saying how many phones these buttons hit.
11	The wall of others	Small live tiles of the other phones, headed `Others · N following`. Clicking a tile swaps that phone onto the stage. A tile click does *not* tap that phone.
12	Status bar	Left: a plain-language summary of the rig. Right: sliders for the stage size and tile size, plus refresh-rate menus. These change how the console draws, never the phones.

The phone stage in detail

This is where all the risk is, so treat it as its own thing.

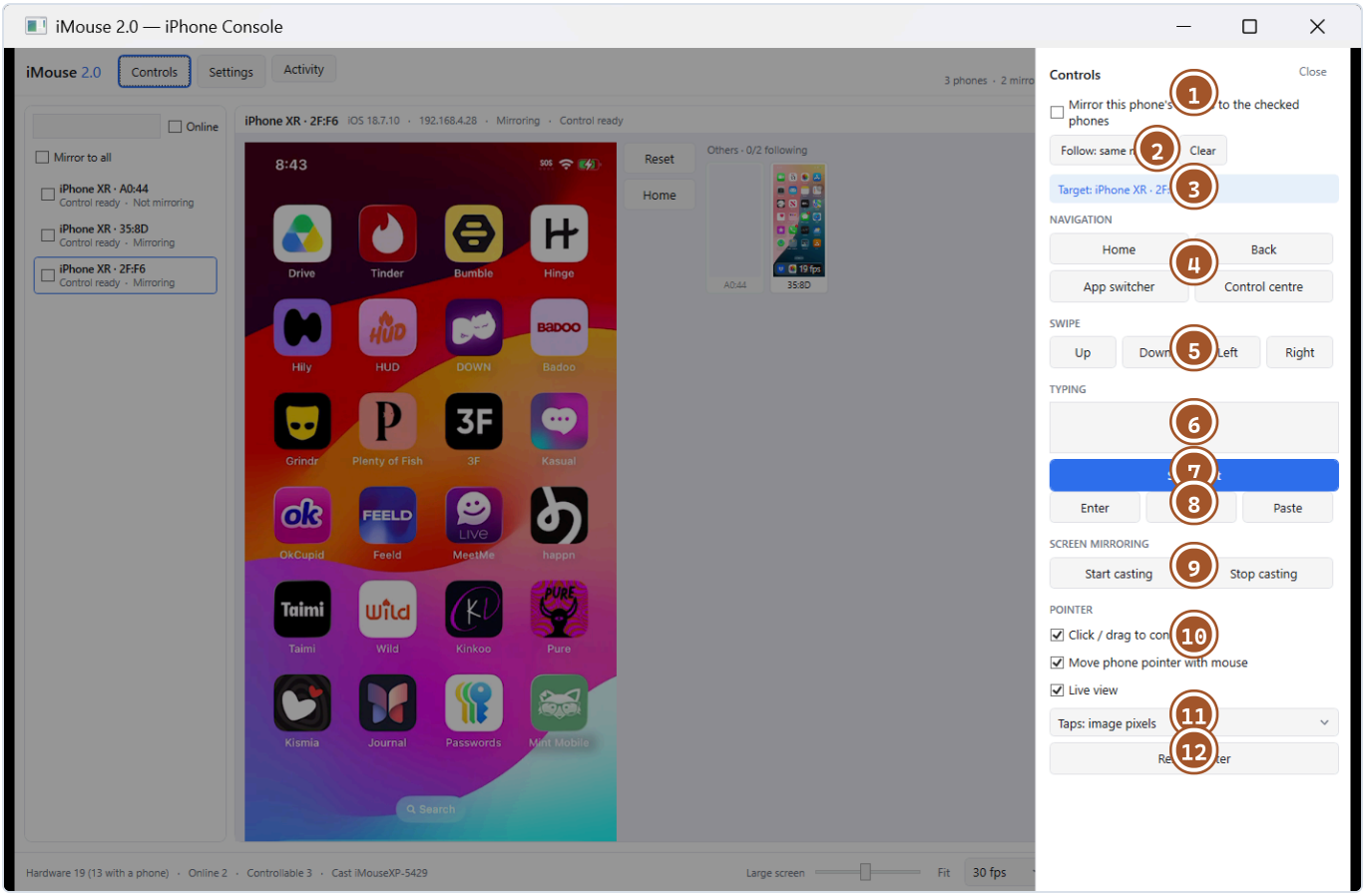
- **A single click is a tap** at the spot you clicked. The console converts your screen position into a phone coordinate; you do not compute anything.
- **A press-drag-release is a swipe** that follows your mouse for the whole gesture. Use it only when a swipe is what you want.
- **A press held still for longer than about a quarter of a second is a long press.** The phone's button goes down while you hold and comes up when you release, so iOS runs its own long-press timer: icon menus, jiggle mode, text selection. The readout changes to `holding` when this has happened. Holding and then moving becomes a drag, which is how you move an icon.
- **Hovering moves the phone's own pointer** to follow your cursor. You will see a small grey dot on the phone. It sends no tap, but park the cursor off the stage when idle so screenshots stay clean.
- **The tap only registers if the picture is fresh.** If the live view is paused or stale, the console refuses the tap and logs `Refresh the preview before controlling the phone.` Turn **Live view** back on and wait for frames.
- **A "Fit" button** in the status bar sizes the stage to the window. A larger stage means more screen pixels per phone pixel, which makes small tap targets easier to hit accurately.

> `TRAP` **The stage is not a safe place to click for any reason other than tapping the phone**
>
> Do not click the stage to bring the window forward, to clear a tooltip, or as a resting place for the cursor. Use the dark border around the stage or the status bar for that.

The three drawers

The buttons that act on phones live in a drawer that slides in from the right when you press **Controls**, **Settings** or **Activity**. Only one drawer is open at a time. A grey overlay covers the rest of the window while it is open; press **Close** or click the overlay to dismiss it.

Controls: the buttons that touch the phone

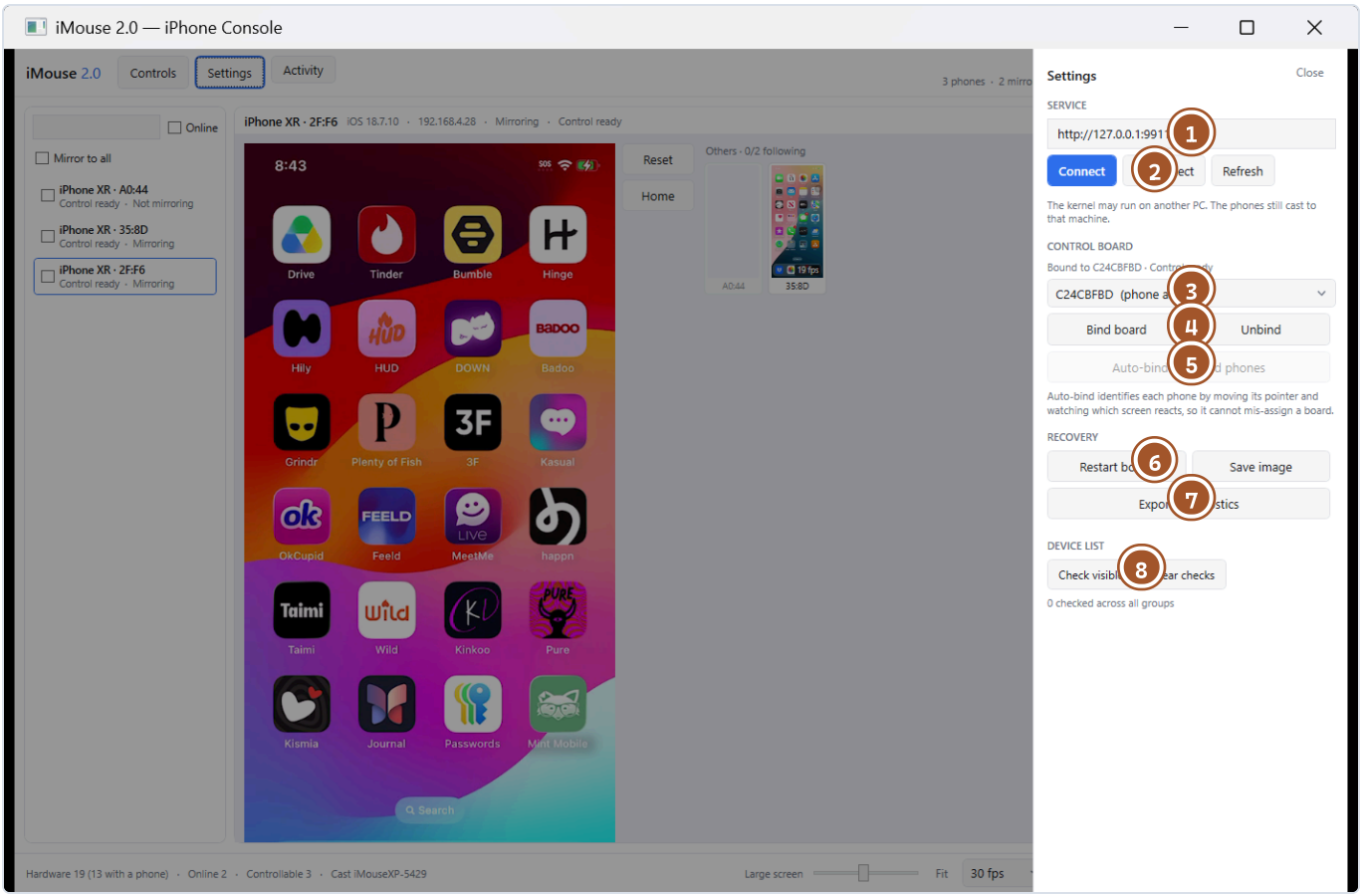


Controls drawer. Everything here acts on the current mirror set: the selected phone, plus any followers if mirroring is on.

#	Control	What it does · how to verify
1	**Mirror this phone's actions to the checked phones**	The real mirror switch. When ticked, every tap, swipe, button and typed string goes to the selected phone and every checked, controllable, same-size phone at once. Verify by the `Master: … → mirroring to N phone(s)` line at 3.
2	**Follow: same model / Clear**	Follow checks every other phone that is the same screen size and controllable, and turns mirroring on. Clear unchecks everything. Neither sends anything to a phone.

#	Control	What it does · how to verify
3	Target line and warning	Blue text names the target: `Target: <phone>` alone, or `Master: <phone> → mirroring to N phone(s)`. Amber text below names any checked phone that will be excluded and why. **Read this before every action.**
4	**Home · Back · App switcher · Control centre**	Navigation. Each sends the matching gesture to the target set. Verify by the phone screen changing.
5	**Up · Down · Left · Right**	Swipes across the middle of the screen. Good for scrolling a feed or changing home pages without a precise drag.
6	Typing box	Type the text you want on the phone *here*, into this box. Nothing is sent until you press **Send text**. English letters, digits and punctuation only.
7	**Send text**	Sends the box's contents to whatever field is focused on the phone. Verify: characters appear in the phone's field, and the log shows `Send text`. Non-ASCII is refused with a log message and nothing is sent.
8	**Enter · Bksp · Paste**	Return key, Backspace, and paste the *phone's* clipboard. These do not clear the typing box.
9	**Start casting · Stop casting**	Turn the phone's screen feed on or off. **Stop casting** blinds you: the picture goes dark. Only use it when instructed. Start casting can take a few seconds to bring the picture back.
10	Pointer checkboxes	**Click / drag to control** (leave ON, or the stage sends nothing), **Move phone pointer with mouse** (ON makes hover move the phone pointer), **Live view** (leave ON, or the picture freezes).
11	**Taps: image pixels** menu	Which coordinate space taps use. Leave on `image pixels`, the proven default. Only change it if taps land offset, and change it back if that does not help.
12	**Reset pointer**	Same as **Reset** in the strip. Re-centres the phone's pointer. The first fix when taps stop landing.

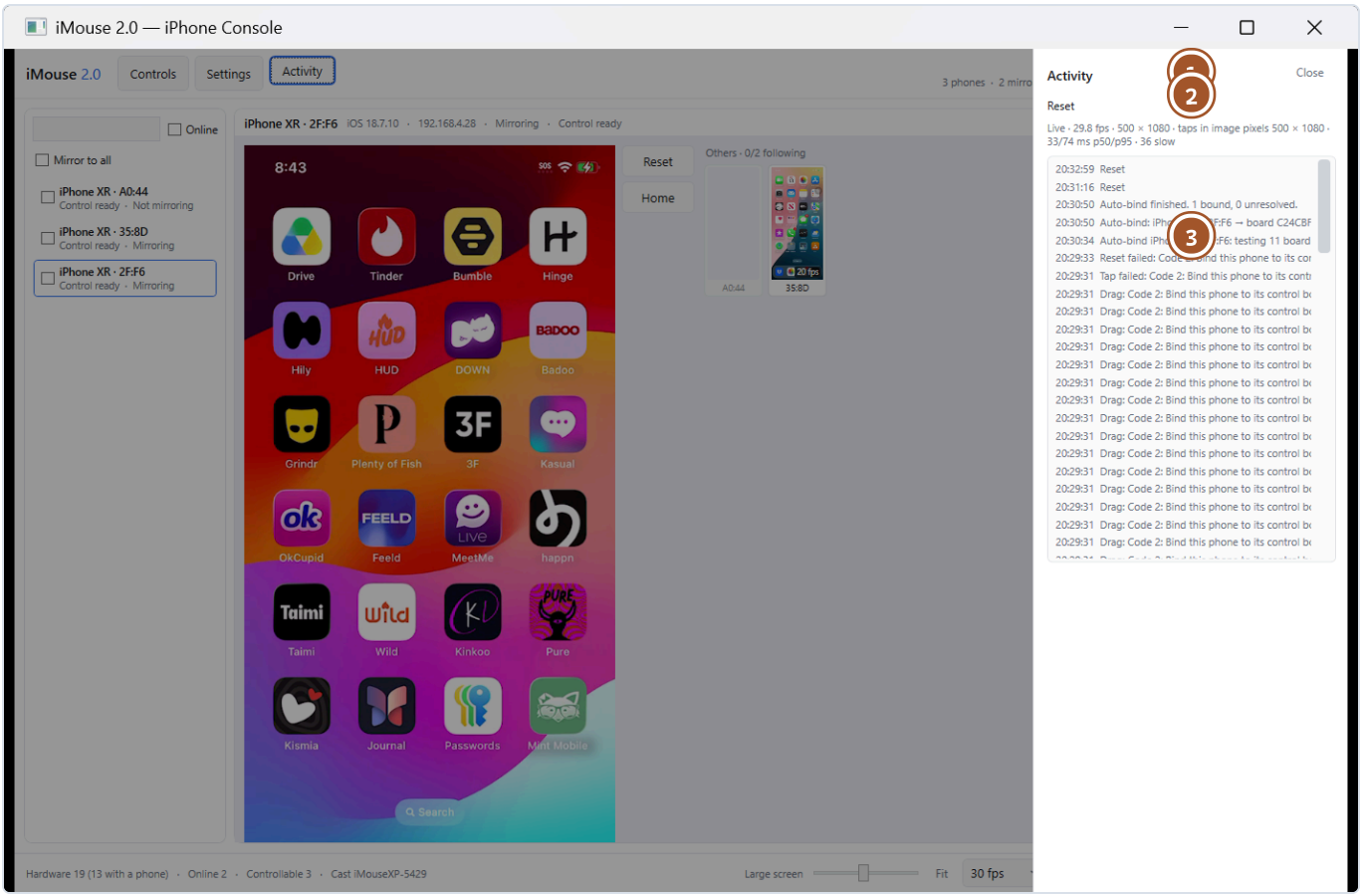
Settings: connection, boards, recovery



Settings drawer. Mostly one-time or recovery controls. An agent should rarely touch this page; several controls here disconnect or unbind a phone.

#	Control	What it does
1	Service address	Where the kernel is. Should read `http://127.0.0.1:9911`. Do not change it.
2	**Connect · Disconnect · Refresh**	Connect/Disconnect the console from the kernel. Refresh re-reads the device list. Refresh is safe; Disconnect stops everything until reconnected.
3	Board picker	Which control board the selected phone is bound to. A support control.
4	**Bind board · Unbind**	Ties or unties a phone and a board. **Unbind** makes a phone uncontrollable. Do not press unless instructed.
5	**Auto-bind unbound phones**	Identifies phones with no board by nudging each pointer and watching which screen reacts. Slow, safe, and a person's job.
6	**Restart board · Save image**	Restart the selected phone's board (a recovery step that drops control briefly). Save the current stage frame to a PNG via a Windows dialog.
7	**Export diagnostics**	Writes a JSON report via a Windows Save dialog. Contains no screenshots or typed text.
8	**Check visible · Clear checks**	Tick every phone currently shown by the filters, or clear all checks. `N checked across all groups` shows the count.

Activity: the log



Activity drawer. Read-only. This is the agent's second source of truth after the picture: it confirms what was sent and reports failures in words.

#	Part	What it tells you
1	Last event	The most recent action in one line, red if it failed.
2	Live status	Frame rate, image size, and tap space, e.g. `Live · 29.8 fps · 500 × 1080 · taps in image pixels · 33/74 ms · 36 slow`. A low fps here means verify slowly.
3	Timestamped log	Every action with its result. `Tap`, `Tap -> 3 phones`, `Home`, `Reset`. Failures name a reason and a code, e.g. `Tap failed: Code 2: Bind this phone to its control board`.

Reading state from the screen

Everything the agent needs to decide is printed on the window in plain words. This section lists those strings and what each one means. Read them from your own screenshot before every phone action. The exact numbers will differ from the examples; the wording will not.

Is the console connected?

Top right of the window.

String	Meaning	Agent action
`Connected to iMouse`	Healthy. The summary line below shows the counts.	Proceed.
`Connecting…`	Still coming up.	Wait, screenshot again.
`Service unavailable · retrying`	The kernel is not answering.	Stop. A person must start the service.
`Disconnected`	The console is not connected.	Stop, or open Settings and press **Connect** if instructed.

The line under the status, and the bottom status bar, both summarise the rig: `3 phones · 2 mirroring · 3 controllable · 19 boards (13 with a phone)` and `Hardware 19 (13 with a phone) · Online 2 · Controllable 3 · Cast iMouseXP-5429`.

Which phone am I about to drive, and can I?

The header above the stage names the selected phone and its two channels. Every device row in the left list says the same two things in miniature.

String on the row or header	Control channel (can you tap it?)
`Control ready` (green)	Yes. Board bound and a phone attached. This is the only state you can drive.
`No phone on board`	No. The board is there but no phone is cabled to it.
`No board bound`	No. The phone has no board assigned.
`Board missing`	No. The bound board is not present.

Second word on the row	Video channel (can you see it?)
`Mirroring`	Yes. The screen is streaming.
`Not mirroring`	No. The stage will be dark for this phone. Control may still work, blind.

> `TRAP` **Two independent lights**
>
> A row reading `Control ready · Not mirroring` can be tapped but not seen. A row reading `No board bound · Mirroring` can be seen but not tapped. Never assume one from the other. If you need to both see and drive a phone, it must say `Control ready` and `Mirroring`.

How many phones will my next action hit?

This is the most important thing to read, because one button can hit the whole rack. Three places tell you, and they agree.

Where	Single phone	Mirroring to many
Green line under **Reset** / **Home**	Absent	`→ 3 phones`
Left panel switch	`Mirror to all` unticked	`Mirror to all  (3 of 3)` ticked
Controls drawer target line	`Target: <phone>`	`Master: <phone> → mirroring to 2 phone(s)`

If any of these shows a count and you intended one phone, stop and clear the mirror set before acting (Section 6). If a checked phone will be silently left out, an amber warning names it:
`1 checked phone(s) have a different screen size and are excluded`, or
`2 checked phone(s) are not mirroring or have no working board`.

Where will my tap land?

As the cursor hovers the stage, the pointer readout at the top right of the stage header shows the phone coordinate: `x 208, y 640 — click to tap here`. If it says `Outside the phone screen`, a click will do nothing useful. You do not need these numbers to act; they are a confirmation that you are over the live area.

Is the picture fresh enough to trust?

The Activity drawer's live line gives the frame rate. Above about 15 fps, a one-second wait after an action is plenty. At `4 fps` or the word `slow`, wait two seconds before you screenshot to verify, or you will read the state from before your action. If it says `Live view paused` or `Live view idle`, the picture is not updating and taps will be refused.

SECTION 6 · STEP BY STEP

Procedures

Common tasks as numbered steps. Each step says what you should see next, marked so. If you do not see it, stop and go to Section 7 rather than repeating the step.

P1 · Select a phone and confirm you can drive it

Do this before any other procedure.

1. Screenshot. Confirm `Connected to iMouse` at top right.
2. In the left list, click the row for the phone you want.
   EXPECT: that phone fills the stage; the header shows its name
3. Read the header's two channel words.
   EXPECT: it says `Control ready` and `Mirroring`
4. Check the mirror indicators (Section 5). For single-phone work, there is no green `→ N phones` line and the drawer says `Target: <phone>`.
   EXPECT: you are pointed at exactly one phone

P2 · Tap something on the phone

1. Do P1 first.
2. Move the cursor over the target on the stage. Read the pointer readout.
   EXPECT: `x …, y … — click to tap here`, not `Outside the phone screen`
3. Single-click.
   EXPECT: the phone reacts within a second; the log shows `Tap`
4. Move the cursor off the stage. Screenshot and confirm the change.

P3 · Type into a field on the phone

1. Tap the field on the phone first (P2), so it has focus and the keyboard is up.
   EXPECT: a cursor blinks in the field on the phone
2. Open **Controls**. Click the typing box. Type your text into it (English letters, digits, punctuation).
   EXPECT: the text appears in the drawer box, not on the phone yet
3. Press **Send text**.
   EXPECT: the characters appear in the phone's field; the log shows `Send text`
4. If you need Return, press **Enter**. For emoji or non-English text, stop: this console cannot send it.

P4 · Navigate: Home, Back, app switcher, swipe

1. Do P1. Confirm single-phone unless you intend the whole set.
2. For Home, press **Home** in the strip beside the stage, or open **Controls** for **Back**, **App switcher**, **Control centre** and the four swipes.

EXPECT: the phone screen changes accordingly; the log names the action

3. To scroll a feed, use **Up**/**Down** in Controls, or drag on the stage. A button swipe is safer than a drag when you do not need a precise distance.

P5 · Swipe or drag precisely on the stage

1. Do P1.
2. Press and hold at the start point on the stage, move to the end point, release. Keep the whole gesture on the stage where you can.
   EXPECT: the content follows your cursor, then settles; the log shows a drag
3. For a plain swipe use one drag action. Separate mouse-down and mouse-up actions are for long presses only (P5b), because a press held longer than about a quarter of a second becomes a real hold on the phone.

P5b · Long press (hold a finger on the screen)

For icon context menus, jiggle mode, text selection, and anything else iOS does on a held touch.

1. Do P1. Confirm single-phone unless you intend the whole set.
2. Move the cursor over the target on the stage and check the pointer readout says `click to tap here`.
3. Press the mouse button down and keep it down. Do not move.
   EXPECT: after about a quarter of a second the readout changes to `x …, y … – holding`, and the phone's button is now down
4. Keep holding until the phone reacts: an icon menu appears, icons start to jiggle, text gets selected. Usually about a second in total.
   EXPECT: the phone shows the long-press result while you are still holding
5. Release the button without moving.
   EXPECT: the phone keeps the menu or jiggle mode open; the log shows `Long press (… ms)`
6. To move an icon instead, keep holding after step 4 and then drag to the destination before releasing. The console carries the hold straight into a drag.
   EXPECT: the log shows `Long press, then drag`

P6 · Do the same thing on several phones at once

Only when you deliberately want the whole group to act together.

1. Select the phone you will treat as the master (P1). It sets the coordinates every follower copies, so it must be the same screen size as them.
2. Either tick **Mirror to all** in the left panel, or in **Controls** tick **Mirror this phone's actions to the checked phones** and tick each follower in the list, or press **Follow: same model**.
   EXPECT: the target line reads `Master: <phone> → mirroring to N phone(s)` and a green `→ N phones` appears by the strip
3. Read the amber warning, if any. It names phones that will be left out. Decide whether that is acceptable.
4. Act once (a tap, a button, a typed string). It goes to all of them in one request.

EXPECT: the log shows `Tap -> N phones`; the wall tiles react

5. **When done, untick the mirror switch** so a later single action does not hit the whole rack by surprise.
   EXPECT: the green line disappears; the target line returns to `Target: <phone>`

P7 · Switch which phone is on the stage

1. Click the phone's row in the left list, or click its small tile on the wall.
   EXPECT: that phone moves to the stage; the previous one moves to the wall
2. Re-read the header and the mirror indicators. Selecting a new master while mirroring is on changes who the followers copy.

P8 · Save a picture of what is on the phone

1. Open **Settings**, press **Save image**.
   EXPECT: a Windows Save dialog opens over the console
2. Accept the default name and folder with `Enter`, or cancel with `Esc`. Do nothing else until the dialog is gone.
   EXPECT: the log shows `Screen image saved.`

Verification and recovery

The phone picture is live, so you can see the result of every action. Use that. Never fire a second phone action before verifying the first. When something does not land, work the ladder below in order rather than repeating the action.

How to verify any action

1. **Wait.** About one second, or two if the live line reads `slow` or under 15 fps.
2. **Screenshot** with the cursor parked off the stage.
3. **Check the picture changed the way you expected.** A tap opened the thing you tapped, the typed text appeared, the app switched.
4. **Read the log** in the Activity drawer as a second opinion. It confirms what was sent, and names any failure in words.

An action can succeed at the console and still not do what you wanted on the phone, so the picture is the real test, not the log alone. Equally, the log catches failures the picture cannot show, such as a rejected coordinate.

The recovery ladder

When a tap or button seems to do nothing, go down this list in order. Stop as soon as the phone responds.

Step	Do	Why
1	Confirm the phone is `Control ready`, not just `Mirroring`.	A phone you can see is not always a phone you can drive. If it is not control ready, no amount of tapping will work.
2	Confirm **Live view** is on and frames are arriving.	Taps are refused on a stale picture, with `Refresh the preview before controlling the phone.` in the log.
3	Press **Reset** once.	A phone can silently stop accepting taps while still moving its pointer. Reset clears that state, and it stays cleared.
4	Retry the tap once.	One retry only. If it still does nothing, do not keep tapping; queued taps can all land at once later.
5	Check the log for a code.	`Code 2: Bind this phone to its control board` and similar tell you exactly what is wrong. See the table below.
6	Open **Settings**, press **Restart board** for that phone, wait a few seconds, then retry.	Resets both ends of the control link. Control drops briefly. This is the last rung an agent should attempt.
7	Stop and raise it to a person.	Beyond a board restart, the fix is physical: a cable, a phone setting, or the vendor console.

What the log's failure messages mean

Log message contains	Meaning	Fix
`Code 2` / `Bind this phone to its control board`	No working board for this phone.	It is not control ready. Pick a phone that is, or have a person bind it.
`Refresh the preview before controlling the phone.`	The picture was stale when you tapped.	Turn on **Live view**, wait for frames, retry.
`Phone orientation changed. Try again.`	The phone rotated between your press and release.	Wait for it to settle, then redo the gesture.
`Keyboard typing supports English letters, numbers and punctuation.`	You tried to send non-ASCII text.	This console cannot. Remove the emoji or accented characters.
`… failed: …` (red)	The kernel rejected the command.	Read the rest of the line; usually control state or connection.

Connection recovery

If the top right reads `Service unavailable · retrying`, the console keeps trying on its own. Do not change the address. If it does not recover in a reasonable time, a person must confirm the vendor service is running. If it reads `Disconnected` and you were told to reconnect, open **Settings** and press **Connect**, then wait for `Connected to iMouse`.

> `TIP` **When in doubt, Reset then look**
>
> The single most useful recovery habit on this rig is: press **Reset**, wait a second, screenshot, and only then decide. It fixes the most common silent fault and it never makes anything worse.

Hazards

Ranked by how much damage a single wrong click does. Everything here has actually happened on this rig or is one click away from happening.

Hazard	What happens	How to avoid it
Clicking the phone picture to focus the window	A real tap lands on the phone at that spot. On a home screen that opens an app; inside an app it can press Send, Buy, Delete or Post.	Focus the window by clicking the dark stage around the picture, the status bar, or the window title bar. Never the picture.
Acting with Mirror to all ticked without meaning to	Every tap, swipe, button and typed string goes to every checked phone in one request.	Read the green `→ N phones` line under Reset and Home before every phone action. If it shows a number you did not intend, untick **Mirror to all** in the left panel.
Typing with the wrong focus	Your keystrokes go into the console's search box or the address box, filtering the device list or changing the service address, while nothing reaches the phone.	Phone typing only ever goes through the **Send text** box and button in the Controls drawer. Before a `type` action, confirm the caret is in that box.
Dragging across the picture on the way somewhere else	Any press-move-release on the picture is a real swipe. A flick across the home screen changes pages; inside an app it scrolls, dismisses or archives.	Use `click` for taps and `drag` only for intended swipes. Use separate mouse-down and mouse-up only for an intended long press (P5b), and remember that any pause with the button down over the picture longer than about a quarter of a second is already a hold on the phone.
Pressing Stop casting	The phone's screen feed ends. The picture goes dark, the phone drops to `Not mirroring`, and you can no longer see what you are doing. Control still works, blind.	Only press it when told to. Recovery is **Start casting** and waiting, and it does not always come back without the vendor console.
Pressing Restart board or Unbind	Control to that phone drops for several seconds, or until a person re-binds it. Unbind leaves the phone visible but untouchable.	These live in the Settings drawer for a reason. Use **Restart board** only as the last rung of recovery in Section 7. Never press **Unbind** unless instructed.

Hazard	What happens	How to avoid it
Touching the vendor's own console window	IMConsole.exe is a separate program that keeps the phones mirroring. Minimising or resizing it repeatedly has left it black and unable to recover; it had to be restarted.	Do not click, move, resize or minimise any window titled iMouse other than `iMouse 2.0 — iPhone Console`.
Changing the service address	The console disconnects from the kernel and shows `Service unavailable · retrying` forever.	Leave **Settings** → **Service** alone. If it is ever wrong, it should read `http://127.0.0.1:9911`.
Repeating a tap that "did nothing"	A phone can silently ignore taps while still showing a perfect picture and still moving its pointer. Repeating the tap sends more taps that will all be ignored, or land all at once when control returns.	One retry at most. Then **Reset**, then the ladder in Section 7.
Trusting the picture on a slow phone	One phone on this rack delivers a new frame about every 220 ms, its twin about every 35 ms. On the slow one, your screenshot can show the state from before your tap.	Wait a full second before verifying. If the top-left tile says `4 fps` or lower, wait two.
Clicking a wall tile expecting a tap	Clicking a small tile does not tap that phone. It swaps that phone into the large view and moves the previous one onto the wall.	Use tile clicks deliberately, to change which phone you are driving. Then re-read the header and the mirror count.
Leaving the cursor over the picture	The phone's pointer sits wherever the cursor rests, and it appears in every screenshot as a small grey dot, which can be mistaken for content.	Park the cursor on the stage or status bar after each action.
Sending non-English text	The console refuses it and logs `Keyboard typing supports English letters, numbers and punctuation.` Nothing reaches the phone.	Keep phone text to ASCII. There is no emoji or accent path in this console.
Native file dialogs	**Save image** and **Export diagnostics** open a Windows Save dialog that covers the console and steals focus.	Complete or cancel the dialog before doing anything else. Its default file name and folder are acceptable; press `Enter` to save or `Esc` to cancel.

Things that look dangerous and are not

- **Hovering over the picture.** Moves the phone's pointer, sends no tap.
- **Clicking a phone in the left list.** Selects which phone is shown large. Sends nothing to any phone.
- **Ticking a phone's checkbox in the left list.** Adds it to the mirror set for later actions. Sends nothing by itself.
- **Opening and closing the drawers.** Purely console UI.
- **Pressing Reset.** Re-centres the phone's pointer. Safe, and the first thing to try when taps stop landing.

- **Pressing Home.** Sends the phone to its home screen. Safe on any app, but it does go to every phone in the mirror set.
- **The sliders and rate boxes in the status bar.** Change how the console draws, not the phones.
- **The theme button.** Repaints the console. Note that every colour in this manual's figures flips when you use it.

SECTION 9 · ONE PAGE TO KEEP OPEN

Quick reference

The whole manual compressed. If the agent reads only one page during a task, this is it.

The five rules

1. A click on the phone picture is a real tap on a real phone. Click the stage only to tap.
2. Read the mirror count before every phone action. One button can hit the whole rack.
3. Your keyboard types into the console. The phone gets text only via the **Send text** box and button, ASCII only.
4. Act, wait about a second, screenshot, verify. Never chain phone actions blind.
5. Moving the mouse over the picture moves the phone's pointer. Park the cursor off the stage when idle.

Where things are

To…	Go to
Pick which phone is on screen	Left list, or a wall tile
Tap or swipe the phone	The large stage directly
Home / Reset	Strip beside the stage
Back, app switcher, swipe buttons, type, cast	**Controls** drawer
Turn mirroring on or off	**Mirror to all** (left), or the **Controls** drawer switch
Connect, boards, restart, save, export	**Settings** drawer
See what was sent and any errors	**Activity** drawer

State words at a glance

You see	It means
`Connected to iMouse`	Good to go
`Service unavailable · retrying`	Stop, get a person
`Control ready` (green)	You can tap this phone
`Mirroring` / `Not mirroring`	You can / cannot see this phone
`→ 3 phones` (green, by the strip)	Your next action hits 3 phones
`Target: <phone>`	Single-phone mode, safe
`Master: … → mirroring to N`	Mirroring mode, N followers
Amber text in the drawer	Some checked phones are being excluded

If a tap does nothing

Is it `Control ready`? Is **Live view** on? Press **Reset**, wait, retry once. Read the log for a code. Then **Restart board**. Then stop and raise it.

Never do these

- Click the stage to focus the window, dismiss something, or rest the cursor.
- Act on the phone without checking the mirror count.
- Repeat a tap more than once when it is not landing.
- Press **Stop casting**, **Unbind**, **Disconnect** or change the service address, unless told to.
- Touch any iMouse window other than `iMouse 2.0 — iPhone Console`.
- Copy a coordinate from this manual. Read your own screen.

The two-channel idea, one last time

Control and video are separate. A phone can be visible and untouchable, or controllable and invisible. Before you rely on a phone, confirm both: it must say `Control ready` and `Mirroring`. Almost every confusing moment on this rig is one channel working while the other does not.